

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

CourseCode:	DSA03	CourseTitle:	NaturalLanguageProcessing
COAssessed:	CO1–Imitation (L1)	Assessment:	AssessmentTool1–Application Based Questions
Weightage:	40% ofCIA		
CO1Statement:	AnalyseNLPmodels,regularexpressionpatterns,finitestateautomata,morphological processing techniques and to interpret language processing. (BL-3)		
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Section/Batch:	A/2024	Date:	13-07-2026

Code of Conduct

I B V Manoj / 192424350 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: BVManoj

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section/Topic	Focus	QNos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Section1:RegularExpressions–ResumeInformationExtraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name. Identify email addresses.

Extract mobile numbers.

Detect technical skills (Python, Java, SQL, Machine Learning, NLP). Extract years of experience.

Generate a structured summary of the candidate's profile.

Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

Problem Statement

Develop a Python program using Regular Expressions to extract candidate information from a resume.

Algorithm

1. Read the resume text.
2. Apply Regular Expressions to extract required information.
3. Search for predefined technical skills.
4. Extract years of experience.
5. Generate a structured summary.
6. Check eligibility criteria.
7. Display the extracted information and eligibility status.

Main Code

```
import re
email=re.findall(r'[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}',resume) phone
= re.findall(r'\b\d{10}\b', resume)
experience=re.search(r'(\d+)\s+years?',resume,re.I)
skills = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
found_skills = [skill for skill in skills if re.search(skill, resume, re.I)]
eligible="Python" in found_skills and int(experience.group(1))>=2
```

Regular Expressions Used

Field	Regex Pattern
Email	[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}
Mobile Number	\b\d{10}\b
Experience	(\d+)\s+years?
Skills	`Python

Sample Input

Name: John David
Email: john@gmail.com
Mobile: 9876543210
Skills: Python, SQL, Machine Learning
Experience: 3 years

Output

Resume Information Extraction

Extracts candidate name, email addresses, mobile numbers, technical skills, and years of experience using regular expressions.

Input Resume Text

☒ Use a sample resume text

Choose a sample resume:

Sample Resume #2 (Name: Jane Doe)

Resume Content

Name: Jane Doe
Contact: jane.doe@company.org, jane_d@work.net
Mobile: 9876543210

PROFESSIONAL BACKGROUND
Data Scientist with 3 years of experience in NLP and Machine Learning.

TECHNICAL SKILLS
Python, SQL, Machine Learning, NLP

Total experience: 3 years

Extract Profile Info

Extracted Profile Details

Candidate Name	Years of Experience
Jane Doe Contact	3 yrs

✓ Candidate is ELIGIBLE (Min 2 years exp & Python skill present)

Email Addresses:

- jane_d@work.net
- jane.doe@company.org

Mobile Numbers:

- 9876543210

Technical Skills Detected:

Python, Sql, Machine Learning, NLP

Show Raw Structured Summary

```
=====
CANDIDATE PROFILE SUMMARY
=====
Name: Jane Doe
Contact: jane_d@work.net, jane.doe@company.org
Email(s): jane_d@work.net, jane.doe@company.org
Mobile Number(s): 9876543210
Technical Skills: Python, Sql, Machine Learning, NLP
Experience: 3 years
Eligibility: ✓ ELIGIBLE
=====
```

OutputExplanation

The program successfully extracted the candidate's personal information using Regular Expressions. It identified the email address, mobile number, technical skills, and years of experience. Since the candidate has Python skill and more than two years of experience, the program marked the candidate as **Eligible**.

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expression that performs the following tasks:

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.

Problem Statement

Develop a Python-based product search system using Regular Expressions that performs exact, prefix, suffix, partial, and case-insensitive searches on product names and generates a report showing the total number of matching products.

Algorithm

1. Store the product names.
2. Accept the search keyword.
3. Perform exact matching.
4. Perform prefix matching.

- 5. Performsuffixmatching.
- 6. Performpartialkeywordmatching.
- 7. Performcase-insensitivesearch.
- 8. Displaymatchingproductsandtotalmatches.

MainCode

```
importre
exact = [p for p in products if re.fullmatch(keyword, p, re.I)]
prefix = [p for p in products if re.match(keyword, p,
re.I)]suffix=[pforpinproductsifre.search(keyword+r'$',p,re.I)]
partial = [p for p in products if re.search(keyword, p, re.I)]
```

RegularExpressionsUsed

SearchType	Pattern
ExactSearch	^keyword\$
PrefixSearch	^keyword
SuffixSearch	keyword\$
PartialSearch	keyword
Case-insensitive	re.IGNORECAS E

SampleInput

Products:
Laptop
LaptopBag
Gaming Laptop
WirelessMouse
Mouse Pad
Keyword:
Laptop

Output

Product Search System

Perform flexible keyword matching on the product catalog using Regular Expressions.

Search Parameters

Enter search keyword:

Laptop

Select Search Type:

Exact Match

Product Catalog

ID	Product Name	Category	Price (\$)
1	Laptop Pro 15	Electronics	1299.99
2	Laptop Stand	Accessories	49.99
3	Wireless Mouse	Accessories	29.99
4	Gaming Laptop Ultra	Electronics	1899.99
5	USB-C Hub	Accessories	39.99
6	Monitor 27 inch	Electronics	449.99
7	Keyboard Mechanical	Accessories	89.99
8	Laptop Bag	Accessories	34.99
9	Wireless Keyboard	Accessories	59.99
10	Laptop Charger	Accessories	24.99

Search Results

Matches Found: 5

ID	Product Name	Category	Price (\$)
1	Laptop Pro 15	Electronics	1299.99
2	Laptop Stand	Accessories	49.99
4	Gaming Laptop Ultra	Electronics	1899.99
8	Laptop Bag	Accessories	34.99
10	Laptop Charger	Accessories	24.99

Search History Report

Show History Details

SEARCH HISTORY REPORT

Total Searches Performed: 20

#	Search Type	Query	Matches
1	Exact	"Laptop"	5
2	Exact	"Laptop"	5
3	Exact	"Laptop"	5
4	Exact	"Laptop"	5
5	Exact	"Laptop"	5
6	Exact	"Laptop"	5
7	Exact	"Laptop"	5
8	Exact	"Laptop"	5

OutputExplanation

The program successfully performed different types of searches using Regular Expressions. It displayed products matching the exact keyword, prefix, suffix, and partial keyword while ignoring case differences. Finally, it generated a report showing the total number of matching products for each search category.

Section3:RegularExpressions–UniversityRegistrationSystem

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.
Validate the institutional email address.
Validate the course code.
Validate the semester information.
Validate the student's mobile number.
Display appropriate validation messages for each field.
Generate a final registration status report indicating whether the registration is successful.

ProblemStatement

Develop a Python program using Regular Expressions to validate student registration details including register number, institutional email address, course code, semester, and mobile number, and generate the final registration status.

Algorithm

1. Read student details.
2. Validate each field using Regular Expressions.
3. Display validation messages.
4. Check whether all validations are successful.
5. Generate the final registration report.

MainCode

```
import re
reg_valid = re.fullmatch(r'\d{9}', reg_no)
email_valid = re.fullmatch(r'[A-Za-z0-9._%+-]+@saveetha\.com', email)
course_valid = re.fullmatch(r'[A-Z]{3}\d{2}', course)
semester_valid = re.fullmatch(r'[1-8]', semester)
mobile_valid = re.fullmatch(r'[6-9]\d{9}', mobile)
```

RegularExpressionsUsed

Field	RegexPattern
RegisterNumber	\d{9}
InstitutionalEmail	[A-Za-z0-9._%+-]+@saveetha.com
CourseCode	[A-Z]{3}\d{2}
Semester	[1-8]
MobileNumber	[6-9]\d{9}

SampleInput

RegisterNumber:192424350
Email:manoj@saveetha.com
Course Code: DSA03 Semester: 3
Mobile:9876543210

Output

University Registration System

Validate student details before course registration using Regular Expressions.

Student Validation Form

☒ Pre-fill with sample student data

Choose a sample student:

Student #1 - REG2024001 (Valid) ▾

Register Number:
REG2024001

Institutional Email Address:
john.smith@university.edu

Course Code:
CS101

Semester:
2nd Semester

Mobile Number:
+91 9876543210

Validation Report

✓ REGISTRATION SUCCESSFUL - All fields validated successfully!

☒ Register Number: REG2024001

☒ Email Address: john.smith@university.edu

☒ Course Code: CS101

☒ Semester: 2nd Semester

☒ Mobile Number: +91 9876543210

Show Raw Registration Report

=====

REGISTRATION STATUS REPORT

=====

Date: 2026-07-13 09:01:18

REGISTRATION INFORMATION:

Register Number: REG2024001

Email: john.smith@university.edu

Course Code: CS101

Semester: 2nd Semester

Mobile Number: +91 9876543210

VALIDATION SUMMARY:

Fields Validated: 5

Passed: 5

Failed: 0

STATUS: ✓ REGISTRATION SUCCESSFUL

=====

OutputExplanation

The program validated each registration field using appropriate Regular Expressions. Since all the entered details matched the required format, the validation was successful, and the system displayed the final registration status as **Registration Successful**.

RubricsforCalculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understandsthe problem and accurately identifies all requirements	Good understanding withminor gaps	Basic understanding; some requirements unclear	Poor understandingor misinterpretation of the problem
Applicationof Concepts	30	Accurately applies appropriate conceptstosolve complex problems	Appliesrelevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Failstoapply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured,and efficient approach	Mostlylogical approachwith minor gaps	Basicapproach with some inconsistencies	Illogical or unclearapproach
Accuracyof Solution	20	Completely correctsolution with proper justification	Mostly correct with minor errors	Partiallycorrect solution	Incorrector incomplete solution
Clarity	10	Clear, well-organized,and properly explained solution	Understandable with minor clarity issues	Limited clarity andexplanation	Poorlypresented and difficult to understand

Q.No./ Criteria	Understandin g of Problem Context	Applicationof Concepts	Problem-Solving Approach	Accuracyof Solution	Clarity	Sub.Total	Total %
1							
2							
3							

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____